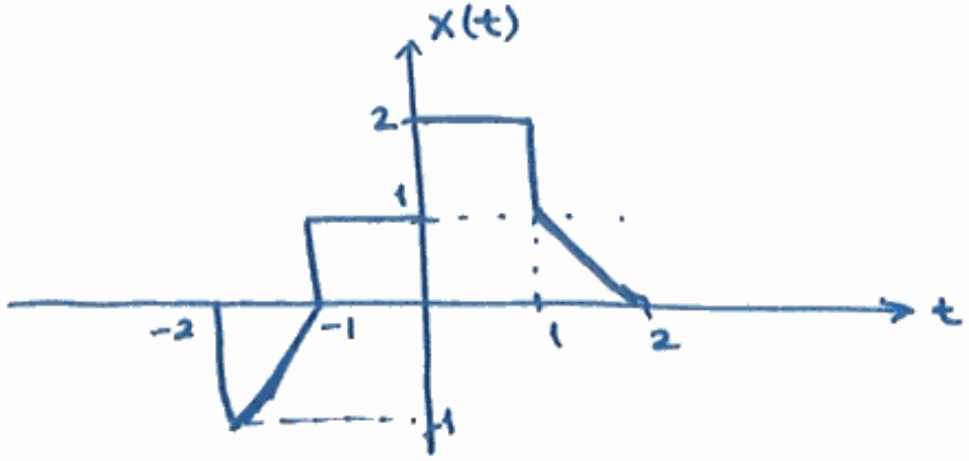


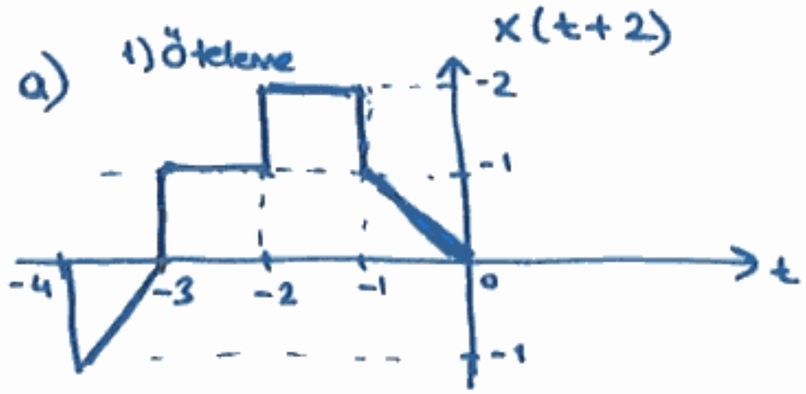
a) $x(2-t)$

b) $x(2t+1)$

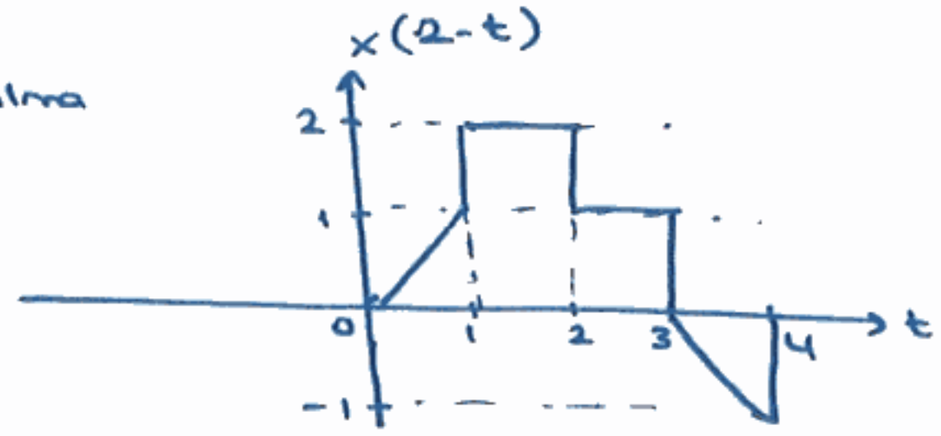
c) $x\left(\frac{t-3}{2}\right)$

1) Öteleme
2) Ters Alma
3) Ölçekleme

a) 1) Öteleme



2) Ters Alma



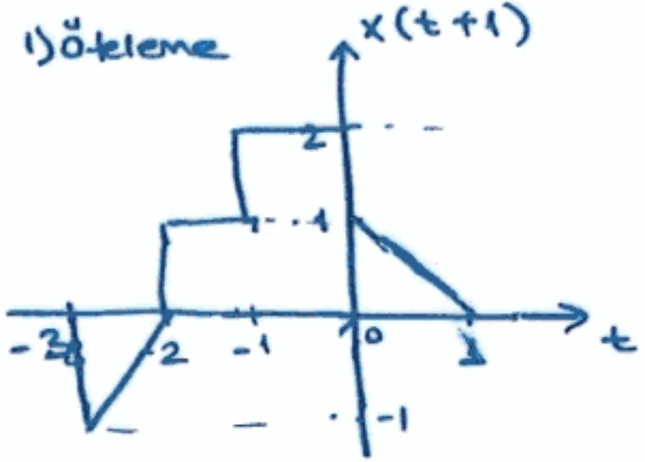
2-t: [-2 2]

t: [4 0]

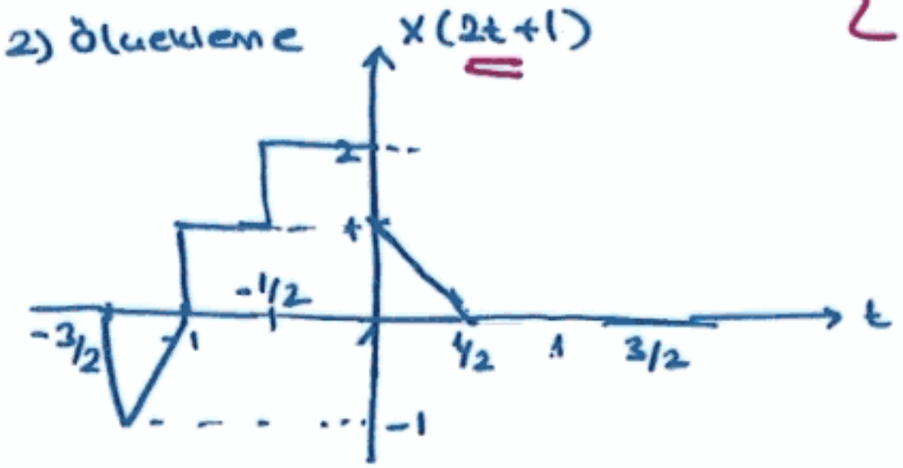
2.yol : $x(t)$, $t:[-2\ \ 2]$

$x(2-t)$, $t:[0\ \ 4]$ ←

b) 1) ötelem



2) ölcsény



t/2 olsódik, 2-vel szorzás

2. yol: x(t), t: [-2 2]

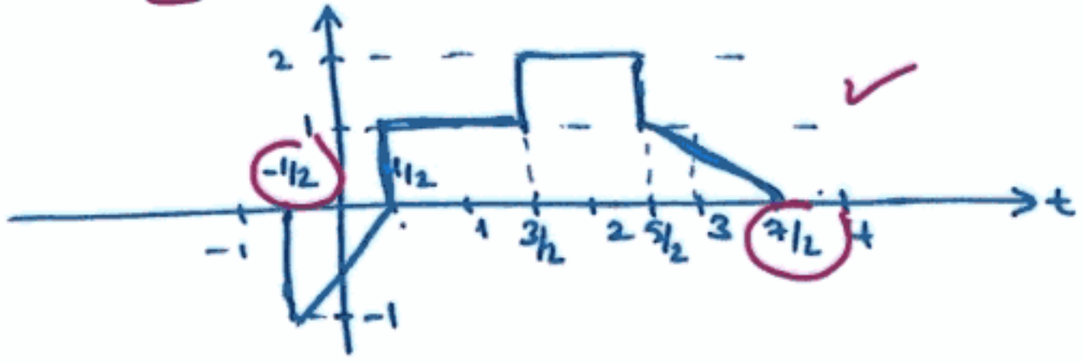
x(2t+1), t: [-3/2 1/2]

2t+1 : [-2 2]

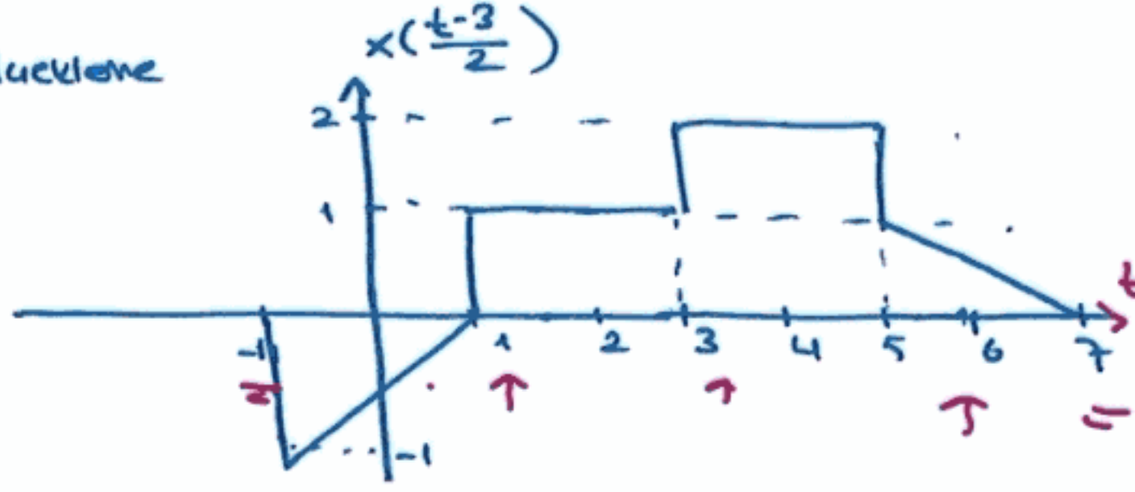
t: [-3/2 1/2]

$x\left(\frac{t-3}{2}\right)$

c) 1) Ötelenme $x(t-3/2)$ →



2) Büzüşme



2.yol : $x(t)$, $t:[-2 \quad 2]$

$x\left(\frac{t-3}{2}\right)$, $t:[-1 \quad 7]$

SORU: $y(t) = t.x(t-2)$ sisteminin a) bellekli b) kararlı c) nedensel d) doğrusal e) zamanda değişmezlik özelliklerine sahip olup olmadığını inceleginiz.

a) Bellekli: $t=2 \rightarrow y(2) = 2.x(0)$ → $x(t)$'nin geçmiş değerine bağlı old. için → Bellekli ✓

b) Kararlılık: $|x(t)| \leq B$ $y(t)$?

$|x(t-2)| \leq B$ ✓

$|t.x(t-2)| \leq \infty \rightarrow -\infty < t < \infty$ old. için.

$|y(t)| < \infty \Rightarrow$ kararlı değildir. ✓

c) Nedensellik: $t=-3 \rightarrow y(-3) = -3.x(-5)$ → Nedensel ✓

d) Doğrusallık: $a.x_1(t) + b.x_2(t) \xrightarrow{?} a.y_1(t) + b.y_2(t)$

$x_1(t) \longrightarrow y_1(t)$ $\quad y_1(t) = t.x_1(t-2)$

$x_2(t) \longrightarrow y_2(t)$ $\quad y_2(t) = t.x_2(t-2)$

$x_3(t) \longrightarrow y_3(t)$

$y_3(t) = t.x_3(t-2) = a.t.x_1(t-2) + b.t.x_2(t-2)$

$y(t) = t \cdot x(t-2)$

$|t \cdot x(t-2)| \leq \infty \rightarrow -\infty < t < \infty$ old. için.

$|y(t)| < \infty \Rightarrow$ kararlı değildir. ✓

c) Nedensellik: $t = -3 \rightarrow y(-3) = -3 \cdot x(-5) \rightarrow$ Nedensel ✓

d) Doğrusallık: $a \cdot x_1(t) + b \cdot x_2(t) \xrightarrow{?} a \cdot y_1(t) + b \cdot y_2(t)$

$x_1(t) \longrightarrow y_1(t)$

$x_2(t) \longrightarrow y_2(t)$

$x_3(t) \longrightarrow y_3(t)$

$x_3(t) = a \cdot x_1(t) + b \cdot x_2(t)$

$y_1(t) = t \cdot x_1(t-2)$

$y_2(t) = t \cdot x_2(t-2)$

- $y_3(t) = t \cdot x_3(t-2) = a \cdot t \cdot x_1(t-2) + b \cdot t \cdot x_2(t-2)$
- $y_3(t) = a \cdot y_1(t) + b \cdot y_2(t)$

$= a \cdot t \cdot x_1(t-2) + b t \cdot x_2(t-2)$

$=$ ✓ Doğrusal. ✓

e) Zamanda değişmezlik: $x(t) \longrightarrow y(t) = t \cdot x(t-2)$

$x(t-t_0) \longrightarrow y(t-t_0)$

$y(t) = t \cdot x(t-2)$ ✓

$x_1(t) = x(t-t_0)$

① $y_1(t) = t \cdot x_1(t-2) = t \cdot x(t-t_0-2)$

② $y(t-t_0) = (t-t_0)\, x(t-t_0-2)$

$\neq$ Zamanda değişir. ✓

$x_1(t) \longrightarrow y_1(t)$

$y_1(t) = t \cdot x_1(t-2)$

$x_1(t) = x(t-t_0)$

$y_1(t) = t\, x(t-t_0-2)$

SORU: Aşağıda giriş-çıkış ilişkisi verilen sistemleri lineerlik bakımından sınıflandırın.

a) $y(t) = 2x(t) - 3$

b) $y(t) = e^{-t} \cdot x(t)$

Lineerlik şartı:

$$x_1(t) \xrightarrow{T[\cdot]} y_1(t)$$
$$x_2(t) \xrightarrow{T[\cdot]} y_2(t)$$

$$T[a \cdot x_1(t) + b \cdot x_2(t)] = a \cdot T[x_1(t)] + b \cdot T[x_2(t)]$$

a) $x(t) \xrightarrow{T[\cdot]} y(t) = 2 \cdot x(t) - 3$

$x_1(t) = a \cdot x(t) \longrightarrow y_1(t) \stackrel{?}{=} T[x_1(t)]$

① $y_1(t) = 2 \cdot a \cdot x(t) - 3$

② $T[x_1(t)] = T[a \cdot x(t)] = a \cdot T[x(t)] = a \cdot y(t) = 2ax(t) - 3a$

$T\{x(t)\} = y(t)$

$x_1(t) \longrightarrow y_1(t) = 2 \cdot x_1(t) - 3$

$x_1(t) = a \cdot x(t) \rightarrow y_1(t) = 2 \cdot a \cdot x(t) - 3$ ✓

$y_1(t) \neq T[x_1(t)]$

↓

lineer değil.

b) $x(t) \longrightarrow y(t) = e^{-t} \cdot x(t)$

$x_2(t) = a \cdot x(t)$ $\quad y_2(t) \stackrel{?}{=} T[x_2(t)]$

$T[x_1(t)] = T[a \cdot x(t)] = a \cdot \underbrace{T[x(t)]}_{y(t)}$

$= a \cdot y(t)$

$= a \cdot 2x(t) - 3a$ ✓

a) $x(t) \xrightarrow{\mathcal{T}[\cdot]} y(t) = 2x(t) - 3$

$x_1(t) = a \cdot x(t) \longrightarrow y_1(t) \stackrel{?}{=} \mathcal{T}[x_1(t)]$

$\mathcal{T}\{x(t)\} = y(t)$

① $y_1(t) = 2 \cdot a \cdot x(t) - 3$

② $\mathcal{T}[x_1(t)] = \mathcal{T}[a \cdot x(t)] = a \cdot \mathcal{T}[x(t)] = a \cdot y(t) = 2ax(t) - 3a$

$x_1(t) \longrightarrow y_1(t) = 2 \cdot x_1(t) - 3$

$x_1(t) = a \cdot x(t) \rightarrow y_1(t) = 2 \cdot a \cdot x(t) - 3$ ✓

$=$

$y_1(t) \neq \mathcal{T}[x_1(t)]$

↓

lineer degil.

b) $x(t) \longrightarrow y(t) = e^{-t} \cdot x(t)$

$x_2(t) = a \cdot x(t)$

$y_2(t) \stackrel{?}{=} \mathcal{T}[x_2(t)]$

$x_2(t) \longrightarrow y_2(t) = e^{-t} \cdot x_2(t)$

$x_2(t) = a \cdot x(t) \longrightarrow y_2(t) = e^{-t} \cdot a \cdot x(t)$ ①

① $y_2(t) = e^{-t} \cdot a x_2(t)$

② $\mathcal{T}[x_2(t)] = \mathcal{T}[a \cdot x(t)] = a \cdot \mathcal{T}[x(t)] = a \cdot e^{-t} \cdot x(t)$

$\mathcal{T}[x_2(t)] = \mathcal{T}[a \cdot x(t)] = a \cdot \mathcal{T}[x(t)] = a \cdot y(t) = a \cdot e^{-t} x(t)$ ②

$y_2(t) = \mathcal{T}[x_2(t)]$

‖

lineer. ✓

SORU: Aşağıda giriş-çıkış ilişkisi verilen sistemleri zamanda değişmezlik bakımından sınıflandırın.

a) $y(t) = 2x(t) + 3$

b) $y(t) = (t+2) \cdot x(t)$

Zamanda Değişmezlik Şartı: $x(t) \longrightarrow y(t)$

$x(t-t_0) \longrightarrow y(t-t_0)$

a) $x(t) \longrightarrow y(t) = 2x(t)+3$

$x_1(t) = x(t-t_0)$

$y_1(t) \overset{?}{=} y(t-t_0)$?

$x(t) \longrightarrow y(t) = 2x(t)+3$

$x_1(t) \longrightarrow y_1(t) = 2x_1(t)+3$

$= 2x(t-t_0)+3$ ①

① $y_1(t) = 2x_1(t)+3 = 2x(t-t_0)+3$

② $y(t-t_0) = 2x(t-t_0)+3$

$y_1(t) = y(t-t_0)$ ✓

zamanda değişmez (time invariant)

b) $x(t) \longrightarrow y(t) = (t+2) \cdot x(t)$ →

$x_2(t) = x(t-t_0)$

$y_2(t) \overset{?}{=} y(t-t_0)$

$y_2(t) = (t+2)\, x_2(t) = (t+2) \cdot x(t-t_0)$

$y(t-t_0) = (t-t_0+2)\, x(t-t_0)$

$y_2(t) \neq y(t-t_0)$

zamanda değişir. (time varion+)

$x_2(t) \longrightarrow y_2(t) =$

b) $x(t) \longrightarrow y(t) = (t+2).x(t)$ $\longrightarrow$

$x_2(t) = x(t-t_0)$ $\qquad y_2(t) \stackrel{?}{=} y(t-t_0)$

① $y_2(t) = (t+2)\,x_2(t) = (t+2).x(t-t_0)$

② $y(t-t_0) = (t-t_0+2)\,x(t-t_0)$

$y_2(t) \neq y(t-t_0)$

zamanda değişir. (time varion+)

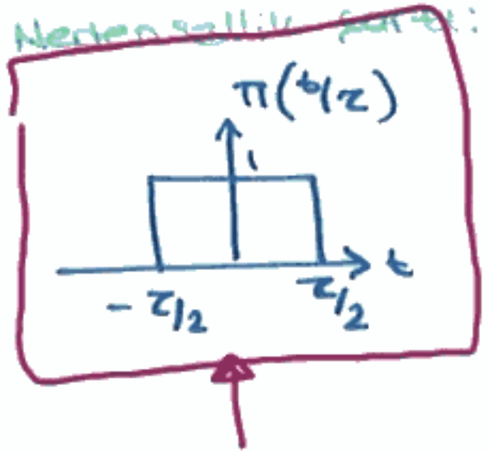
$x_2(t) \longrightarrow y_2(t) = (t+2).x_2(t)$

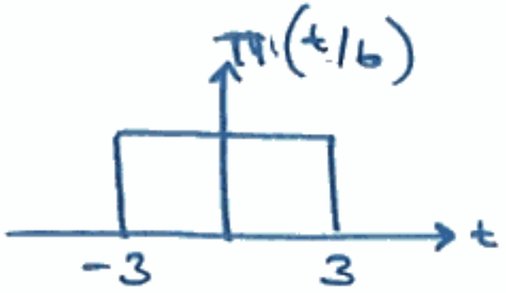
$y_2(t) = (t+2)\,x(t-t_0)$ ①

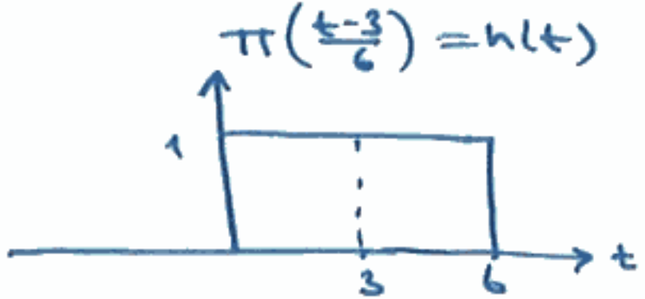
LZD

SORU: $h(t) = \Pi\left(\frac{t-3}{6}\right)$ Birim dürtü cevabı verilen LTI sistem nedensel midir?

Nedensellik şartı: $t<0$, $h(t)=0$ ise sistem nedenseldir!

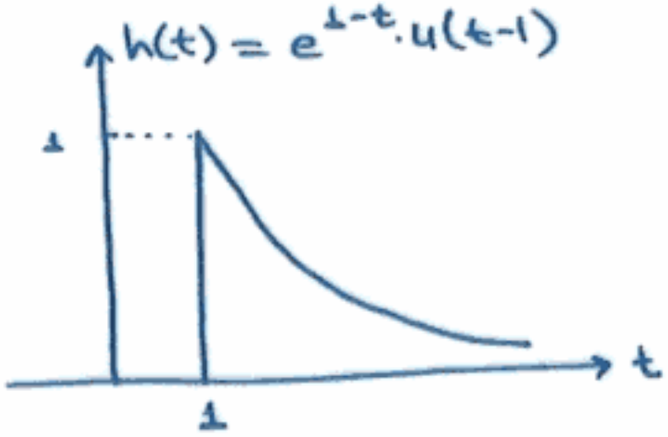


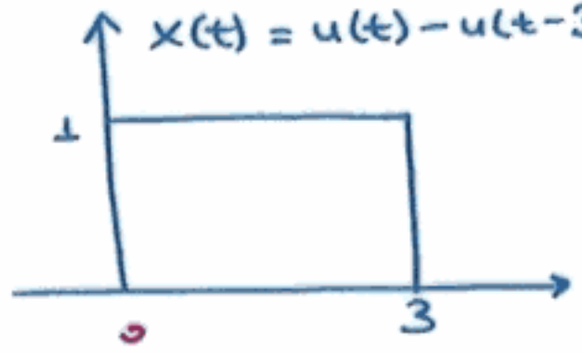




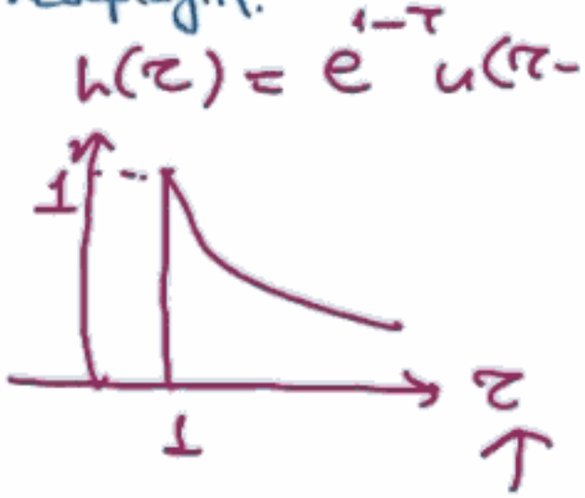
$t<0$ iken $h(t)=0$ olduğu için sistem Nedenseldir!

rından LTI sistemin cıkışını konvolüsyon kullanarak hesaplayın.

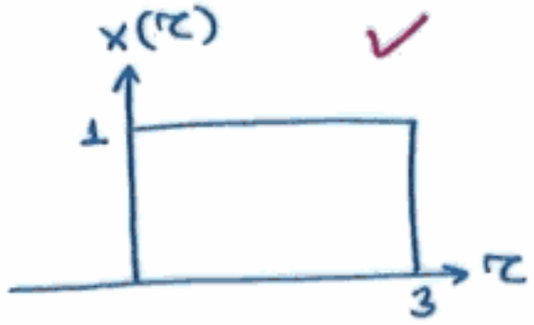




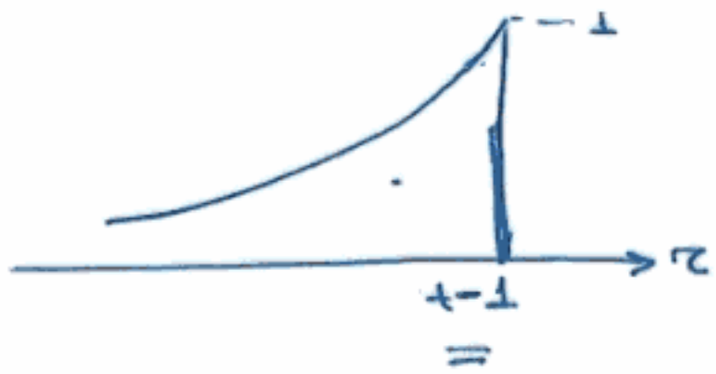
=



$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$



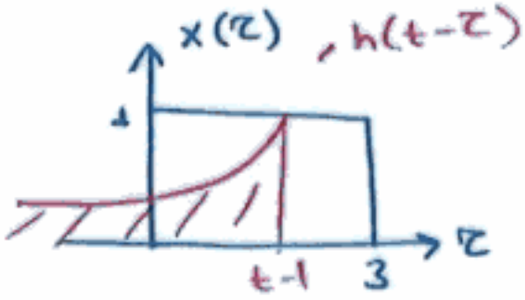
$h(t-\tau) = e^{-(t-\tau-1)} \cdot u(t-\tau-1)$



=

$t-1 < 0$ ise $y(t) = 0$

$0 \leq t-1 < 3 \rightarrow 1 \leq t < 4$

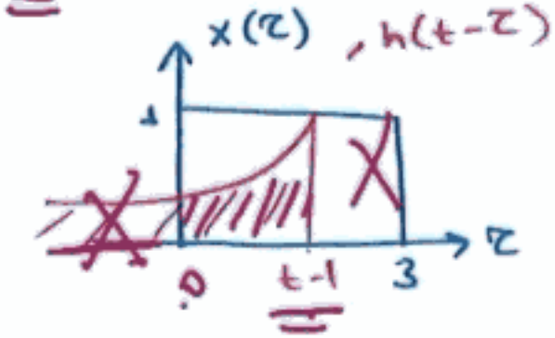


$$y(t) = \int_0^{t-1} e^{-(t-\tau-1)}\, d\tau = e^{-(t-1)} \cdot \int_0^{t-1} e^{\tau}\, d\tau$$

$$= e^{-(t-1)} \left(e^{t-1} - 1\right) = 1 - e^{-(t-1)}$$

$t-1<0$ ise $y(t)=0$ ✓

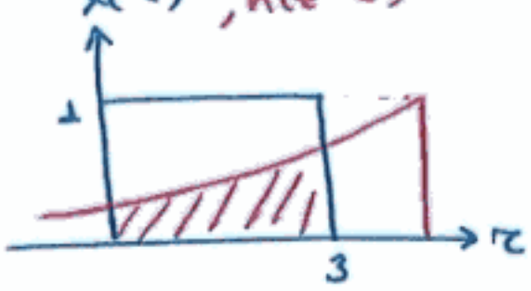
$0\le t-1<3 \rightarrow \boxed{1\le t<4}$



$$y(t)=\int_0^{t-1} 1\cdot e^{-(t-\tau-1)}\,d\tau = e^{-(t-1)}\int_0^{t-1} e^{\tau}\,d\tau = e^{-(t-1)}\cdot e^{\tau}\Big|_0^{t-1}$$

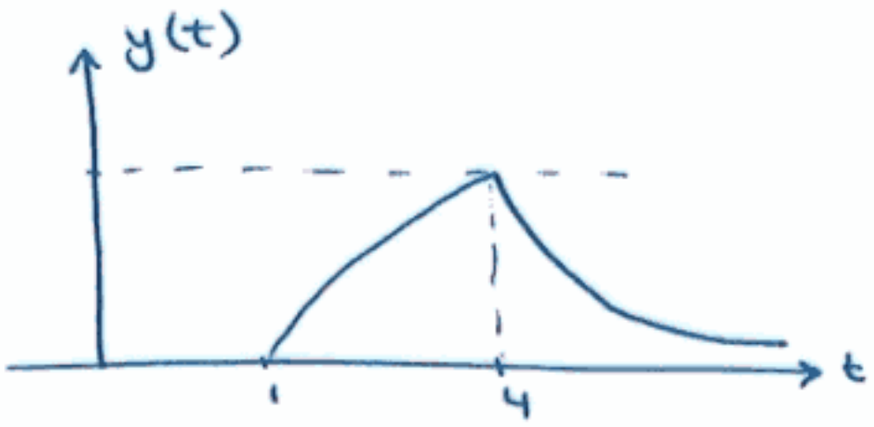
$$= e^{-(t-1)}\left(e^{t-1}-1\right) = 1-e^{-(t-1)}, \quad 1\le t<4$$

$t-1>3 \rightarrow t>4$



$$y(t)=\int_0^{3} e^{-(t-\tau-1)}\,d\tau = e^{-(t-1)}\int_0^{3} e^{\tau}\,d\tau = e^{-(t-1)}\cdot\left(e^{3}-1\right)$$

$$= e^{-t}\left(e^{4}-e^{1}\right)$$

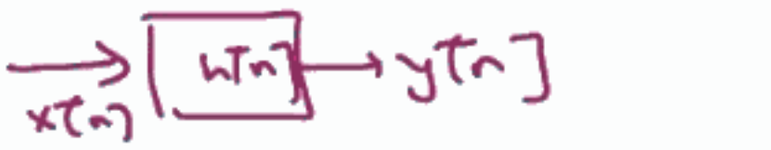


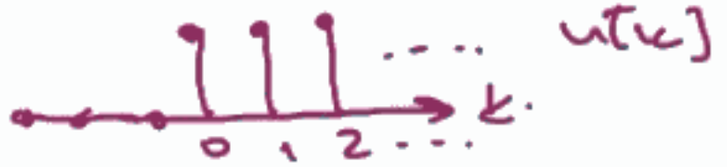
$$y(t)=\begin{cases} 0, & t<1 \\ 1-e^{-(t-1)}, & 1\le t<4 \\ e^{-t}\left(e^{4}-e^{1}\right), & t\ge 4 \end{cases}$$

SORU: Ayrık zamanda aşağıdaki giriş ve dürtü cevapları için konvolüsyon hesaplayın ve grafiğini çizin.

a) $x[n] = (0.8)^n \cdot u[n]$ ve $h[n] = (0.9)^n \cdot u[n]$

b) $x[n] = \delta[n] - \delta[n-2]$ ve $h[n] = u[n]$







a) $$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = \sum_{k=-\infty}^{\infty} \underbrace{(0.8)^k \cdot u[k]}_{x[k]} \cdot \underbrace{(0.9)^{n-k} \cdot u[n-k]}_{h[n-k]}$$

$$= (0.9)^n \sum_{k=0}^{n} \left(\frac{0.8}{0.9}\right)^k \quad , n \geq 0$$

$$= (0.9)^n \cdot \frac{1 - \left(\frac{0.8}{0.9}\right)^{n+1}}{1 - \frac{8}{9}} \quad , n \geq 0$$

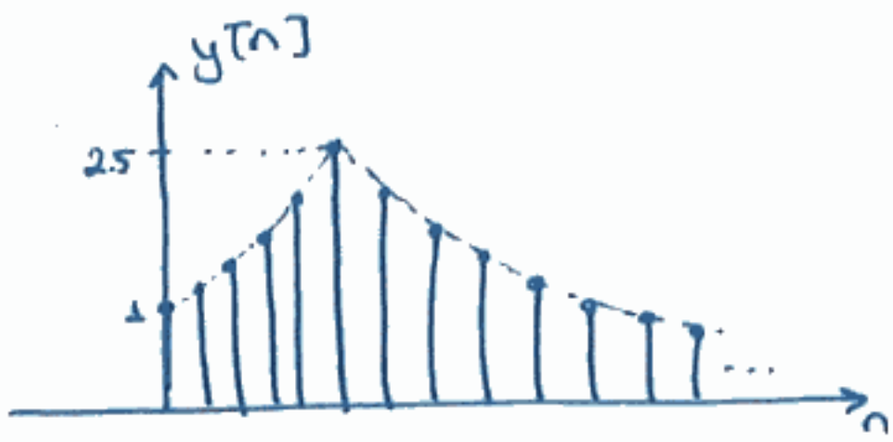
$$= \frac{(0.9)^n \left[(0.9)^{n+1} - (0.8)^{n+1}\right]}{\frac{1}{9} \cdot (0.9)^n \cdot (0.9)} \quad , n \geq 0$$

$$= \frac{\left((0.9)^{n+1} - (0.8)^{n+1}\right)}{0.1} \cdot u[n]$$

Hatırlatma:

$$\sum_{k=0}^{n} \alpha^k = \begin{cases} \frac{1-\alpha^{n+1}}{1-\alpha}, & \alpha \neq 1 \\ n+1, & \alpha = 1 \end{cases}$$





$= (0.9)^n \cdot \frac{\left(\frac{8}{9}\right)^n}{1-\frac{8}{9}}$, $n \geq 0$

$n-2$ $n-1$ n

$x[n] * h[n] = \sum_k x[k]\, h[n-k]$

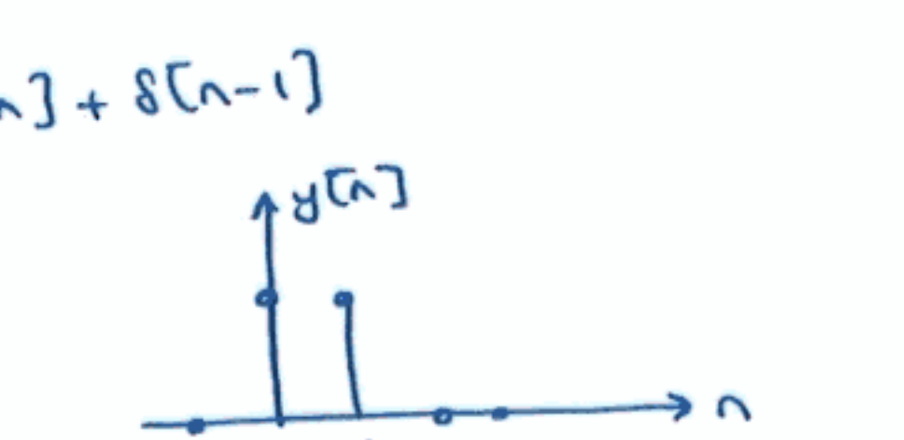
$\delta[k] * u[n] = \sum_k \delta[k]\, u[n-k]$

$= \left(\frac{(0.9)^n \left[(0.9)^{n+1} - (0.8)^{n+1}\right]}{\frac{1}{9} \cdot (0.9)^n \cdot (0.9)} \right)$, $n \geq 0$

$n \geq 0$

$u[n]$

$$y[n] = \frac{\left((0.9)^{n+1} - (0.8)^{n+1}\right)}{0.1} \cdot u[n]$$



$x[n] = \delta[n] - \delta[n-2]$ $\quad h[n] = u[n]$

$x[k]$ $\quad h[n-k]$

b) $y[n] = \sum_{k=\langle \infty \rangle} x[k]\, h[n-k] = \sum_k \left(\delta[k] - \delta[k-2]\right) \cdot u[n-k]$

$\delta[k] * u[k]$

$= \sum_k \delta[k]\, u[n-k] - \sum_k \delta[k-2]\, u[n]$

$\sum_k = \sum_{k=\langle \infty \rangle}$

$= u[n] - u[n-2] = \delta[n] + \delta[n-1]$

$x[n] \cdot \delta[n-n_0] = x[n_0]\, \delta[n-n_0]$

$x[n] * \delta[n-n_0] = x[n-n_0]$

$$\begin{cases} x[n] = \delta[n] - \delta[n-2] \\ h[n] = u[n] \end{cases}$$

b) $y[n] = \sum_{k=<\infty>} x[k]\,h[n-k] = \sum_{k} \big(\delta[k] - \delta[k-2]\big)\cdot u[n-k]$

($x[k]$, $h[n-k]$)

$x[n] * u[n]$

$= \sum_{k} \delta[k]\,u[n-k] - \sum_{k} \delta[k-2]\,u[n-k]$

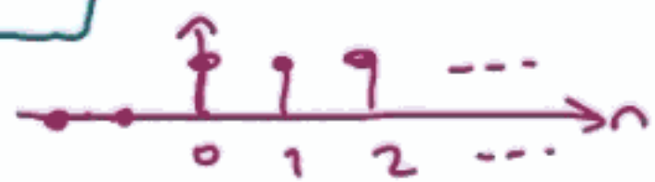
$\sum_{k} \delta[k]\,u[n-k] = \delta[n] * u[n]$

$= u[n]$

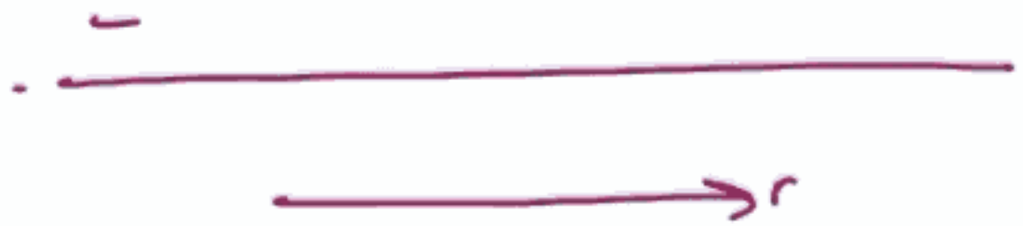
$= u[n] - u[n-2] = \delta[n] + \delta[n-1]$

$u[n-2]$

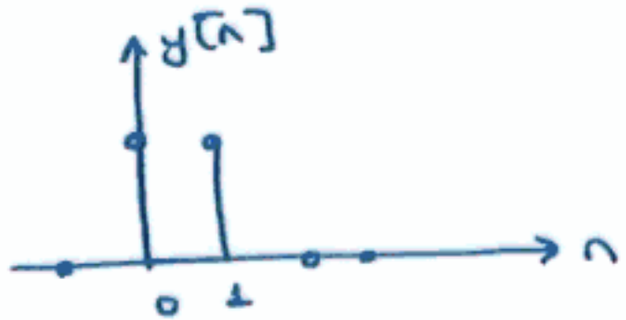
$$x[n] * \delta[n-n_0] = x[n-n_0]$$











SORU:

$x[n] = u[n-3]$

$h[n] = -\delta[n] + \delta[n+2]$

$x[n] \rightarrow \boxed{h[n]} \rightarrow y[n]$

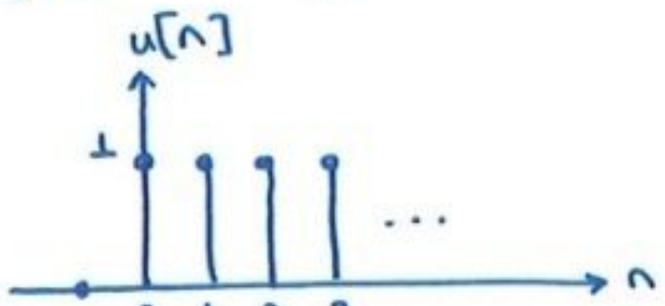
ise $y[n] = ?$

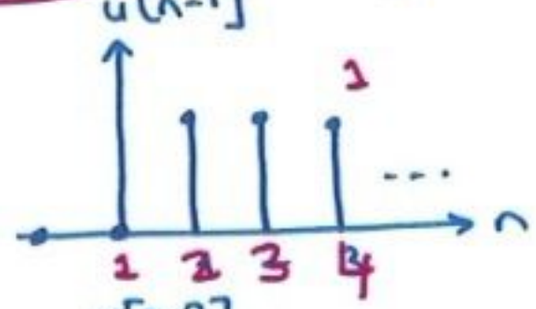
$x(t) * \delta(t-t_0) = x(t-t_0)$

$x[n] * \delta[n-n_0] = x[n-n_0]$

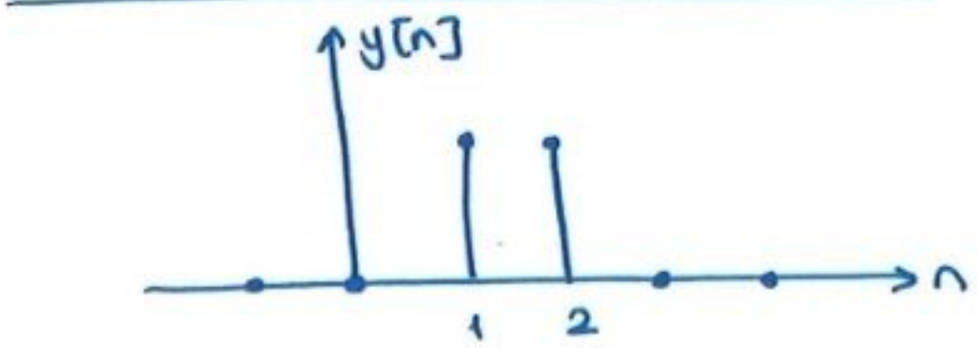
$y[n] = x[n] * h[n] = \overbrace{u[n-3]}^{x[n]} * \overbrace{\left(-\delta[n] + \delta[n+2]\right)}^{h[n]} = -u[n-3] + u[n-1]$

$u[n-3] * \delta[n+2] = u[n+2-3] = u[n-1]$









$\Rightarrow y[n] = \delta[n-1] + \delta[n-2]$

SORU: Seri bağlanmış iki LTI sistemi düşünün. Bunlardan ilkinin dürtü cevabı $h_1[n] = \sin(8n)$ ve ikincisinin dürtü cevabı $h_2[n] = a^n u[n]$, $|a|<1$ ve giriş işareti $x[n] = \delta[n] - a.\delta[n-1]$ 'dir. Çıkış işareti $y[n]$'yi bulunuz.



$$y[n] = (x[n] * h_1[n]) * h_2[n]$$

$$= (x[n] * h_2[n]) * h_1[n]$$

$x[n]\delta[n-n_0] = x[n_0].\delta[n-n_0]$

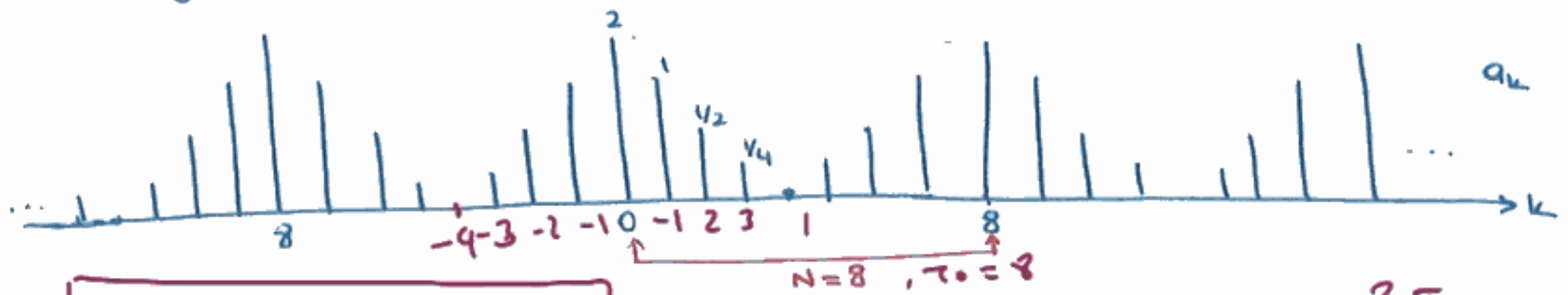
$$x[n] * h_2[n] = (\delta[n] - a\,\delta[n-1]) * a^n.u[n]$$

$$= a^n.u[n] - a.a^{n-1}.u[n-1] = a^n\,(u[n] - u[n-1]) = a^n.\delta[n] = \delta[n]$$

(a^n; $u[n]-u[n-1] = \delta[n]$; $n=0$)

$$y[n] = \delta[n] * h_1[n] = h_1[n] = \sin 8n \quad \checkmark$$

SORU: Aşağıda Fourier serisi katsayıları verilen $x[n]$ işaretini bulunuz.



$$x[n] = \sum_{k=\langle N \rangle} a_k \, e^{-jk\omega_0 n}$$

$$\omega_0 = 2\pi f_0 = \frac{2\pi}{N} \qquad \frac{2\pi}{T_0}$$

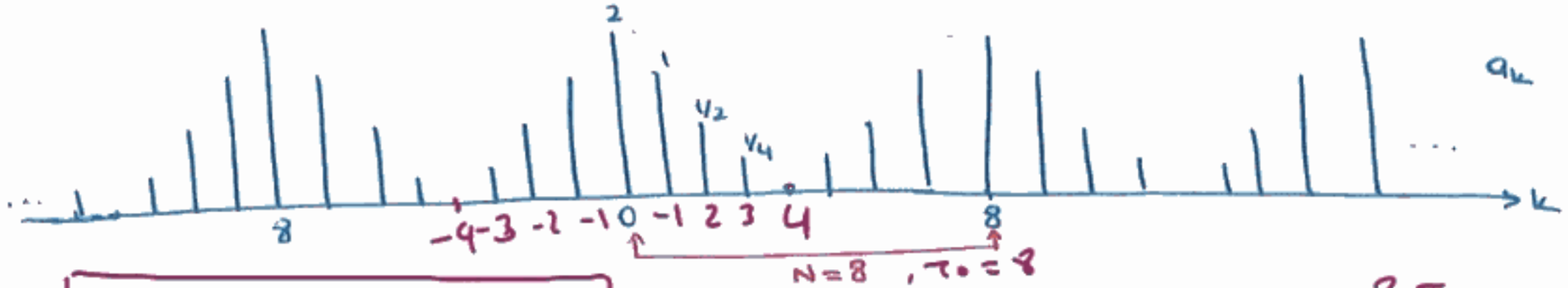
$$\omega_0 = \frac{2\pi}{8} = \frac{\pi}{4} \checkmark$$

$$x[n] = \sum_{k=-4}^{4} a_k \cdot e^{jk\omega_0 n} = \underbrace{2}_{k=0} + \underbrace{1 \cdot \left[e^{-j\omega_0 n} + e^{j\omega_0 n} \right]}_{a_1 = a_{-1}} + \underbrace{\frac{1}{2}\left[e^{-j2\omega_0 n} + e^{j2\omega_0 n} \right]}_{a_2 = a_{-2}} + \underbrace{\frac{1}{4}\left[e^{-j3\omega_0 n} + e^{j3\omega_0 n} \right]}_{a_3 = a_{-3}}$$

(a_0)

$$= 2 + 2\cos(\omega_0 n) + \cos(2\omega_0 n) + \frac{1}{2}\cos(3\omega_0 n)$$

$$= 2 + 2 \cdot \cos\left(\frac{\pi n}{4}\right) + \cos\left(\frac{\pi n}{2}\right) + \frac{1}{2}\cos\left(\frac{3\pi n}{2}\right)$$



$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n}$$

$$\omega_0 = 2\pi f_0 = \frac{2\pi}{N} \qquad \frac{2\pi}{T_0}$$

$$\omega_0 = \frac{2\pi}{8} = \frac{\pi}{4} \checkmark$$

$$x[n] = \sum_{k=-4}^{4} a_k e^{jk\omega_0 n} = \underbrace{2}_{a_0} + \underbrace{1}_{k=0}\cdot\underbrace{\left[e^{-j\omega_0 n} + e^{j\omega_0 n}\right]}_{2\cos(\omega_0 n)} + \underbrace{\frac{1}{2}\left[e^{-j2\omega_0 n} + e^{j2\omega_0 n}\right]}_{a_2 = a_{-2}} + \underbrace{\frac{1}{4}\left[e^{-j3\omega_0 n} + e^{j3\omega_0 n}\right]}_{a_3 = a_{-3}}$$

$a_1 = a_{-1}$

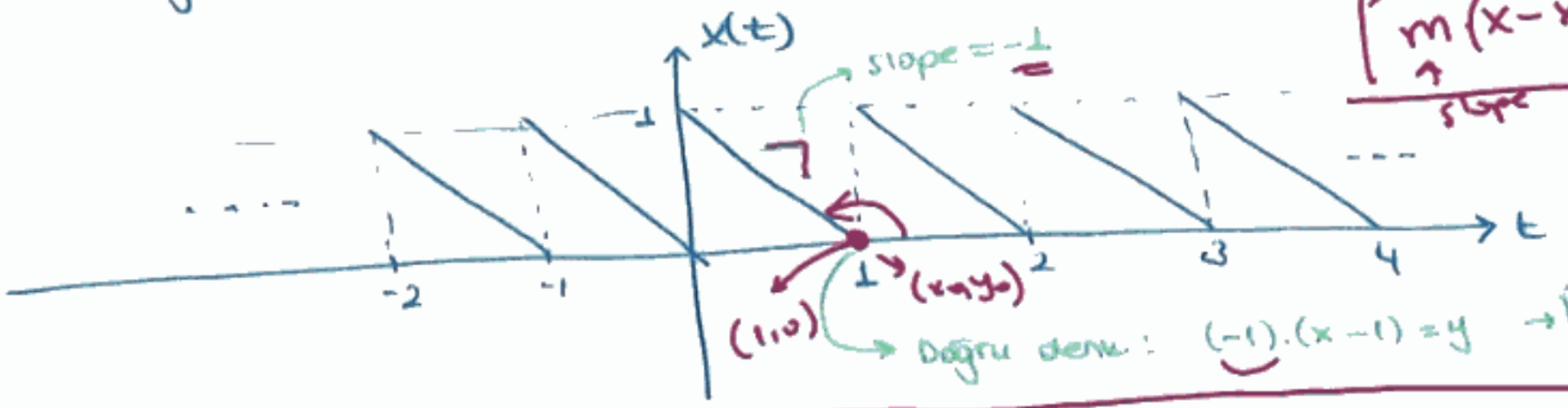
$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$= 2 + 2\cos(\omega_0 n) + \cos(2\omega_0 n) + \frac{1}{2}\cos(3\omega_0 n)$$

$$x[n] = 2 + 2\cdot\cos\left(\frac{\pi n}{4}\right) + \cos\left(\frac{\pi n}{2}\right) + \frac{1}{2}\cos\left(\frac{3\pi n}{2}\right)$$

Soru: Aşağıda verilen periyodik işaretin Fourier seri katsayılarını bulunuz.



$$m(x-x_0) = y-y_0$$

(m: slope)

periyot: $T_0 = 1$ $\rightarrow \omega_0 = \frac{2\pi}{T_0} = 2\pi$ ✓

Üstel seri Açılımı: $x(t) = \sum_{k=-\infty}^{\infty} x_k \cdot e^{jk\omega_0 t}$

$$x_k = \frac{1}{T_0}\int_{T_0} x(t)\cdot e^{-jk\omega_0 t}\,dt$$

$$x_k = \int_0^1 (1-t)\cdot e^{-jk2\pi t}\,dt = \underbrace{\int_0^1 e^{-jk2\pi t}\,dt}_{I_1} - \underbrace{\int_0^1 t\cdot e^{-jk2\pi t}\,dt}_{I_2} = I_1 - I_2$$

$$I_1 = \int_0^1 e^{-jk2\pi t}\,dt = \frac{-1}{jk2\pi} e^{-jk2\pi t}\Big|_0^1 = \frac{-1}{jk2\pi}\left[e^{-j2\pi k} - 1\right] = \frac{1}{j2\pi k}\left(1 - e^{-j2\pi k}\right)$$

$$I_2 = \int_0^1 t\cdot e^{-jk2\pi t}\,dt$$

Kısmi integrasyon: $\int u\,dv = u.v - \int v\,du$

$$x_K = \int_0^1 (1-t)\cdot e^{-jK2\pi t}\,dt = \underbrace{\int_0^1 e^{-jK2\pi t}\,dt}_{I_1} - \underbrace{\int_0^1 t\cdot e^{-jK2\pi t}\,dt}_{I_2} = I_1 - I_2 \;\checkmark$$

$$\int e^{-at}\,dt = -\frac{1}{a}e^{-at}$$

$$I_1 = \int_0^1 e^{-jK2\pi t}\,dt = \frac{-1}{jK2\pi}\,e^{-jK2\pi t}\Big|_0^1 = \frac{-1}{jK2\pi}\left[e^{-j2\pi K} - 1\right] = \frac{1}{j2\pi K}\left(1 - e^{-j2\pi K}\right)\checkmark$$

$$I_2 = \int_0^1 t\cdot e^{-jK2\pi t}\,dt$$

Kısmi integrasyon: $\int u\,dv = u.v - \int v\,du$

$u = t$ $\quad du = du$

$e^{-jK2\pi t}\,dt = dv$ $\quad \frac{-1}{jK2\pi}\,e^{-jK2\pi t} = v$

$$I_2 = t\cdot\frac{-1}{jK2\pi}\,e^{-jK2\pi t}\Big|_0^1 + \int_0^1 \frac{1}{jK2\pi}\,e^{-jK2\pi t}\,dt$$

$$= \frac{-1}{jK2\pi}\,e^{-jK2\pi} - 0 + \frac{1}{jK2\pi}\cdot\frac{-1}{jK2\pi}\cdot e^{-jK2\pi t}\Big|_0^1$$

$$x(t) = \sum_{K=-\infty}^{\infty} \frac{1}{j2\pi K}\cdot e^{-jK2\pi t}$$

$$I_2 = \int_0^1 \underbrace{t}_{u} \cdot \underbrace{e^{-jk2\pi t}\,dt}_{dv}$$

$u = t$ $\quad du = dt$

$e^{-jk2\pi t}\,dt = dv$ $\quad \frac{-1}{jk2\pi} e^{-jk2\pi t} = v$

$$I_2 = t \cdot \frac{-1}{jk2\pi} e^{-jk2\pi t}\Big|_0^1 + \int_0^1 \frac{1}{jk2\pi} e^{-jk2\pi t}\,dt$$

$$= \frac{-1}{jk2\pi} e^{-jk2\pi} - 0 + \frac{1}{jk2\pi} \cdot \frac{-1}{jk2\pi} \cdot e^{-jk2\pi t}\Big|_0^1$$

$\frac{1}{j2\pi k}\left(1 - e^{-j2\pi k}\right)$

$$= \frac{-1}{j2\pi k} \cdot e^{-j2\pi k} + \frac{1}{j^2(2\pi k)^2}\left(1 - e^{-j2\pi k}\right)$$

$$x_k = I_1 - I_2 = \frac{1}{j2\pi k} - \frac{1}{j2\pi k} e^{-j2\pi k} + \frac{1}{j2\pi k} e^{-j2\pi k} + \frac{1}{(2\pi k)^2}\left(1 - \underbrace{e^{-j2\pi k}}_{=1}\right) = \frac{1}{j2\pi k}$$

$$x(t) = \sum_{k=-\infty}^{\infty} \frac{1}{j2\pi k} \cdot e^{-jk2\pi t}$$

$$I_2 = \int_0^1 \underbrace{t}_{u} \cdot \underbrace{e^{-jk2\pi t}\,dt}_{dv}$$

$u = t$ $\quad du = dt$

$e^{-jk2\pi t}\,dt = dv$ $\quad \frac{-1}{jk2\pi}\,e^{-jk2\pi t} = v$

$$I_2 = t \cdot \frac{-1}{jk2\pi} e^{-jk2\pi t}\Big|_0^1 + \int_0^1 \frac{1}{jk2\pi} e^{-jk2\pi t}\,dt$$

$$= \frac{-1}{jk2\pi} e^{-jk2\pi} - 0 + \frac{1}{jk2\pi} \cdot \frac{-1}{jk2\pi} \cdot e^{-jk2\pi t}\Big|_0^1$$

$$\frac{1}{jk2\pi}\left(1 - e^{-j2\pi k}\right)$$

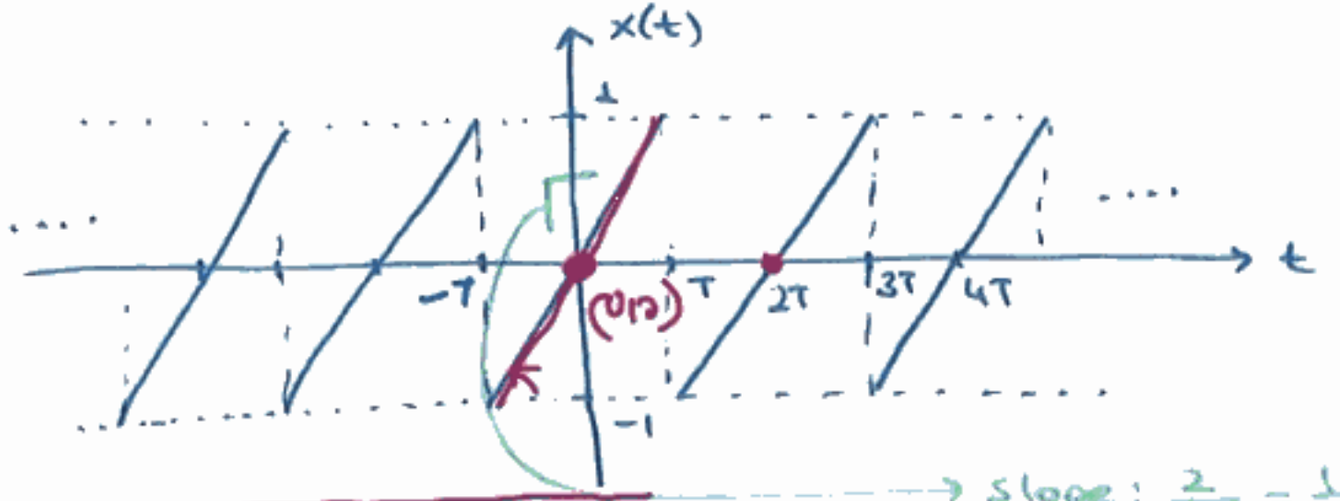
$$I_2 = \frac{-1}{j2\pi k} \cdot e^{-j2\pi k} + \frac{1}{j^2(2\pi k)^2}\left(1 - e^{-j2\pi k}\right)$$

$$x_k = I_1 - I_2 = \frac{1}{j2\pi k} - \cancel{\frac{1}{j2\pi k} e^{-j2\pi k}} + \cancel{\frac{1}{j2\pi k} e^{-j2\pi k}} + \frac{1}{(2\pi k)^2}\left(1 - \underbrace{e^{-j2\pi k}}_{=1}\right) = \frac{1}{j2\pi k}$$

$$x(t) = \sum_{k=-\infty}^{\infty} \frac{1}{j2\pi k} \cdot e^{-jk2\pi t}$$

$e^{j\theta} = \cos\theta + j\sin\theta$

SORU: Aşağıda verilen periyodik işaretin Fourier serisi açılımını bulunuz.



slope: $\frac{2}{2T} = \frac{1}{T}$ ✓ D.D: $\frac{1}{T}x = y$ ✓

$T_0 = 2T$, $\omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{2T} = \frac{\pi}{T}$ ✓

$x(t) = \frac{t}{T}$

$$x_K = \frac{1}{T_0}\int_{T_0} x(t)\cdot e^{-jk\omega_0 t}\,dt$$ ✓

$$= \frac{1}{2T}\int_{-T}^{T} \frac{t}{T}\cdot e^{-jk\frac{\pi}{T}\cdot t}\,dt = \frac{1}{2T^2}\int_{-T}^{T} t\cdot e^{-jk\frac{\pi}{T}\cdot t}\,dt$$

* Kısmi İntegrasyon!

$u = t$, $du = dt$

$e^{-jk\frac{\pi}{T}\cdot t}\,dt = dv$, $\frac{-1}{jk\frac{\pi}{T}} e^{-jk\frac{\pi}{T}t} = v$

$$= \frac{1}{2\tau^2}\left\{ t \cdot \frac{-1}{jk\frac{\pi}{\tau}} \cdot e^{-jk\frac{\pi}{\tau}t}\Big|_{-\tau}^{\tau} + \int_{-\tau}^{\tau} \frac{1}{jk\frac{\pi}{\tau}} e^{-jk\frac{\pi}{\tau}t}\, dt \right\}$$

$$= \frac{1}{2\tau^2}\left\{ \frac{-1}{jk\frac{\pi}{\tau}} \cdot \left[\tau \cdot e^{-jk\pi} + \tau \cdot e^{jk\pi}\right] + \frac{1}{jk\frac{\pi}{\tau}} \cdot \frac{-1}{jk\frac{\pi}{\tau}t} \cdot e^{-jk\frac{\pi}{\tau}t}\Big|_{-\tau}^{\tau} \right\}$$

$$= \frac{1}{2\tau^2}\left\{ \frac{j\tau^2}{\pi k}\left[e^{-j\pi k} + e^{j\pi k}\right] + \frac{1}{(k\cdot\frac{\pi}{\tau})^2}\left[e^{-j\pi k} - e^{j\pi k}\right] \right\}$$

$e^{j\pi n} = (-1)^n$

$e^{-j\pi n} = (-1)^n$

$$= \frac{j}{2\pi k} \cdot 2 \cdot (-1)^k + \frac{1}{2(\pi k)^2} \cdot 0 = \frac{j}{\pi k} \cdot (-1)^k = \frac{(-1)^{k+1/2}}{\pi k}$$

$$x(t) = \sum_{k=-\infty}^{\infty} \frac{(-1)^{k+1/2}}{\pi k} e^{jk\frac{\pi}{\tau}\cdot t}$$

$$= \frac{1}{2\tau^2}\left\{ t \cdot \frac{-1}{jk\frac{\pi}{\tau}} \cdot e^{-jk\frac{\pi}{\tau}t}\Big|_{-\tau}^{\tau} + \int_{-\tau}^{\tau} \frac{1}{jk\frac{\pi}{\tau}} e^{-jk\frac{\pi}{\tau}t}\, dt \right\}$$

$e^{-j2\pi k} = 1$

$$= \frac{1}{2\tau^2}\left\{ \frac{-1}{jk\frac{\pi}{\tau}} \cdot \left[\tau \cdot e^{-jk\pi} + \tau \cdot e^{jk\pi}\right] + \frac{1}{jk\frac{\pi}{\tau}} \cdot \frac{-1}{jk\frac{\pi}{\tau}t} \cdot e^{-jk\frac{\pi}{\tau}t}\Big|_{-\tau}^{\tau} \right\}$$

$e^{j\pi n} = (-1)^n$

$e^{-j\pi n} = (-1)^n$

$$= \frac{1}{2\tau^2}\left\{ \frac{j\tau^2}{\pi k}\left[e^{-j\pi k} + e^{j\pi k}\right] + \frac{1}{(k\cdot\frac{\pi}{\tau})^2}\left[e^{-j\pi k} - e^{j\pi k}\right] \right\}$$

$$= \frac{j}{2\pi k} \cdot 2 \cdot (-1)^k + \frac{1}{2(\pi k)^2} \cdot 0 = \frac{j}{\pi k} \cdot (-1)^k = \frac{(-1)^{k+1/2}}{\pi k}$$

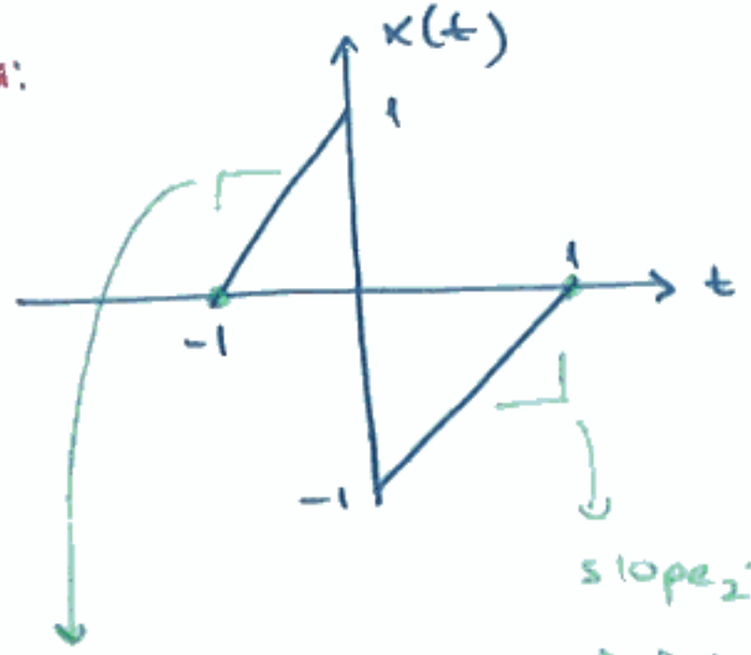
$$x(t) = \sum_{k=-\infty}^{\infty} \frac{(-1)^{k+1/2}}{\pi k} e^{jk\frac{\pi}{\tau}\cdot t}$$

$(e^{j\pi})^n$

$e^{j\pi} = \cos\pi + j\sin\pi = -1 \longrightarrow (-1)^n$

$e^{-j\pi} = -1 \longrightarrow (e^{-j\pi})^n = (-1)^n$

Soru:



$slope_1 = +1$

D.D: $1.(x+1) = y$

$slope_2 = +1$

D.D: $1.(x-1) = y$

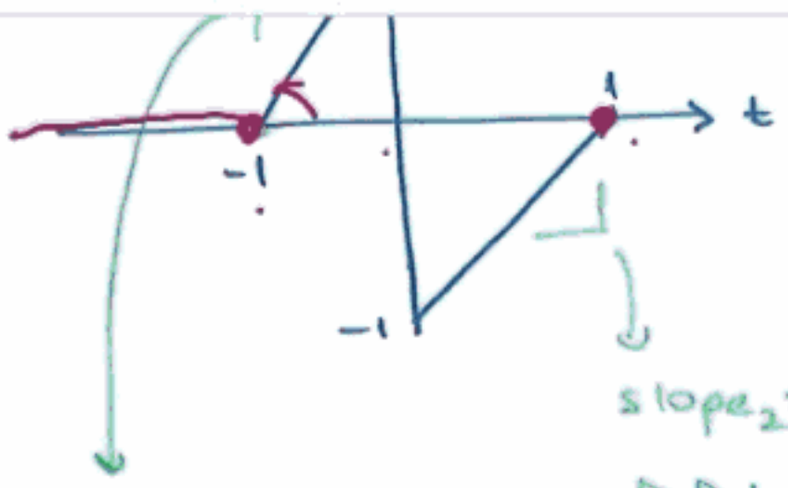
x(t) işaretinin FT'nu bulunuz

$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$$

$$X(f) = \int_{-1}^{0} (t+1)\cdot e^{-j2\pi ft}\, dt + \int_{0}^{1} (t-1)\cdot e^{-j2\pi ft}\, dt$$

$$= \int_{-1}^{0} t\cdot e^{-j2\pi ft}\, dt + \int_{-1}^{0} e^{-j2\pi ft}\, dt + \int_{0}^{1} t\cdot e^{-j2\pi ft}\, dt - \int_{0}^{1} e^{-j2\pi ft}\, dt$$

$$= \frac{j}{\pi f}\left[1 - \operatorname{sinc}(2f)\right]$$



$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt \quad \checkmark$$

$slope_2 = +1$

D.D: $1.(x-1) = y \quad x(t) = t-1$

$slope_1 = +1$

D.D: $1.(x+1) = y \rightarrow x(t) = t+1$

$$X(f) = \int_{-1}^{0} (t+1)\cdot e^{-j2\pi ft}\, dt + \int_{0}^{1} (t-1)\cdot e^{-j2\pi ft}\, dt$$

$$\int t \cdot e^{-at}\, dt$$

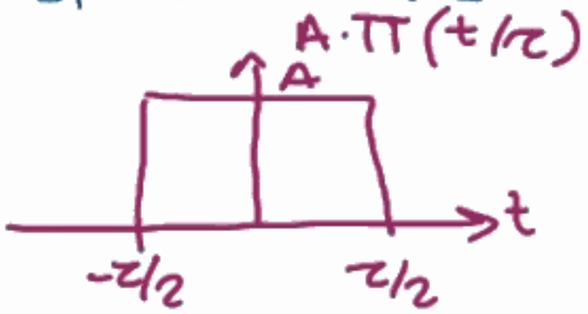
$$= \int_{-1}^{0} t\cdot e^{-j2\pi ft}\, dt + \int_{-1}^{0} e^{-j2\pi ft}\, dt + \int_{0}^{1} t\cdot e^{-j2\pi ft}\, dt - \int_{0}^{1} e^{-j2\pi ft}\, dt$$

$$= \frac{j}{\pi f}\left[1 - \operatorname{sinc}(2f)\right] \quad \checkmark$$

$$\frac{\sin(\pi x)}{\pi x} = \operatorname{sinc}(x)$$

SORU: $x(t) = \Pi(2t+3) + \Pi(3t-2)$ işaretinin Fourier transformunu bulunuz.

$x(t) \xrightarrow{F} X(f)$



Hatırlatma: $A\cdot\Pi(t/\tau) \xrightarrow{F} A\tau\,\mathrm{sinc}(f\tau)$ ←

$x(t-t_0) \xrightarrow{F} X(f)\,e^{-j2\pi f t_0}$ ←

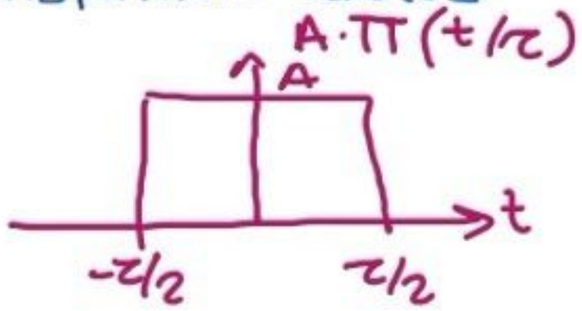
$$\Pi\left(\frac{t+3/2}{1/2}\right) \longrightarrow \frac{1}{2}\cdot\mathrm{sinc}\left(f/2\right)\cdot e^{-j2\pi f(-3/2)}$$

$$\Pi\left(\frac{t-2/3}{1/3}\right) \longrightarrow \frac{1}{3}\,\mathrm{sinc}\left(\frac{f}{3}\right)e^{-j2\pi f\cdot 2/3}$$

$$X(f) = \frac{1}{2}\,\mathrm{sinc}\left(\frac{f}{2}\right)\cdot e^{+j2\pi f\cdot 3/2} + \frac{1}{3}\,\mathrm{sinc}\left(\frac{f}{3}\right)e^{-j2\pi f\cdot 2/3}$$

SORU: $x(t) = \Pi(2t+3) + \Pi(3t-2)$ işaretinin Fourier transformunu bulunuz.

$x(t) \xrightarrow{F} X(f)$



Hatırlatma: $A\cdot\Pi(t/\tau) \xrightarrow{F} A\cdot\tau\,\mathrm{sinc}(f\tau)$ ←

$x(t-t_0) \xrightarrow{F} X(f)\,e^{-j2\pi f t_0}$

$\Pi(2t+3) = \Pi\left(\frac{t+3/2}{1/2}\right) \longrightarrow \frac{1}{2}\cdot\mathrm{sinc}\left(f/2\right)\cdot e^{-j2\pi f(-3/2)}$

$2(t+3/2)$ $\quad t_0 = -3/2$ $\quad \tau\cdot\frac{1}{2}$ $\quad f\cdot\tau = \frac{f}{2}$

$\Pi\left(\frac{t-2/3}{1/3}\right) \longrightarrow \frac{1}{3}\,\mathrm{sinc}\left(\frac{f}{3}\right) e^{-j2\pi f\cdot 2/3}$

$$X(f) = \frac{1}{2}\,\mathrm{sinc}\left(\frac{f}{2}\right)\cdot e^{+j2\pi f\cdot 3/2} + \frac{1}{3}\,\mathrm{sinc}\left(\frac{f}{3}\right) e^{-j2\pi f\cdot 2/3}$$

Soru: $x(t) = \overbrace{4\cdot\Pi\left(t/4\right)}^{x'}\cdot\cos(2\pi f_0 t)$ işaretinin fourier transformu bulunuz.

*Modülasyon

Hatırlatma: $x(t)\cdot\cos(2\pi f_0 t) \longrightarrow \frac{1}{2}\left[X(f-f_0) + X(f+f_0)\right]$ ✓

$$x'(t) = 4\cdot\Pi\left(t/4\right) \longrightarrow \hat{X}(f) = 16\cdot\mathrm{sinc}(f\cdot 4)$$

$$X(f) = \frac{1}{2}\cdot 16\cdot\mathrm{sinc}\left(4(f-f_0)\right) + \frac{1}{2}\cdot 16\cdot\mathrm{sinc}\left(4(f+f_0)\right)$$

$$= 8\,\mathrm{sinc}\left(4(f-f_0)\right) + 8\,\mathrm{sinc}\left(4(f+f_0)\right)$$

Soru: $x(t) = \overbrace{4\cdot\Pi\left(t/4\right)}^{\hat{x}(t)}\cdot\cos(2\pi f_0 t)$ işaretinin fourier transformunu bulunuz.

#Modülasyon

$\hat{x}(t)$'nin işaretinin fourier transformu bulunur.

$$\rightarrow X(f) = \frac{1}{2}\left[\hat{X}(f-f_0) + \hat{X}(f+f_0)\right]$$

Hatırlatma: $x(t)\cdot\cos(2\pi f_0 t) \longrightarrow \frac{1}{2}\left[X(f-f_0) + X(f+f_0)\right]$ ✓

$$\hat{x}(t) = 4\cdot\Pi\left(t/4\right) \longrightarrow \hat{X}(f) = 16\cdot\text{sinc}(f\cdot 4)$$

$A=4\quad \tau=4$ $\qquad = \qquad \frac{\tau}{A\tau}$

$$X(f) = \frac{1}{2}\cdot 16\cdot\underbrace{\text{sinc}(4(f-f_0))}_{X(f-f_0)} + \frac{1}{2}\cdot 16\cdot\underbrace{\text{sinc}(4(f+f_0))}_{X(f+f_0)}$$

$$= 8\,\text{sinc}(4(f-f_0)) + 8\,\text{sinc}(4(f+f_0))$$ ✓

$$A\cdot\Pi(t/\tau) \longrightarrow A\tau\,\text{sinc}(f\tau)$$

$x(t) = \frac{1}{1+t^2}$ Find the Fourier Transform.

$$e^{-\alpha|t|} \longrightarrow \frac{2\alpha}{\alpha^2 + (2\pi f)^2} = \frac{2\alpha}{4\pi^2} \cdot \frac{1}{\frac{\alpha^2}{4\pi^2} + f^2}$$

$$\left.\begin{array}{l} x(t) \longrightarrow X(f) \\ X(t) \longrightarrow x(-f) \end{array}\right\} \text{Duality}$$

$$X(t) = \frac{1}{1+t^2} \xrightarrow{F} x(-f)$$

$$\frac{2\alpha}{4\pi^2} \quad \frac{1}{\frac{\alpha^2}{4\pi^2} + t^2} \xrightarrow{F} e^{-\alpha|-f|} = e^{-\alpha|f|}$$

$$\frac{1}{\frac{\alpha^2}{4\pi^2} + t^2} \longrightarrow \frac{4\pi^2}{2\alpha} \cdot e^{-\alpha|f|}$$

$$\frac{\alpha^2}{4\pi^2} = 1 \qquad \alpha^2 = 4\pi^2 \rightarrow \underline{\alpha = 2\pi}$$

$$\frac{1}{1+t^2} \xrightarrow{F} \pi \cdot e^{-2\pi|f|}$$

14) $x(t) = \frac{1}{1+t^2} \stackrel{?}{=} Y(t)$ Find the Fourier Transform. $X(f) = \int_{T_0} x(t) e^{\cdots} dt$

$e^{-\alpha|t|} \xrightarrow{\mathcal{F}} \frac{2\alpha}{\alpha^2 + (2\pi f)^2} = \frac{2\alpha}{4\pi^2} \cdot \frac{1}{\frac{\alpha^2}{4\pi^2} + f^2}$

$e^{-\alpha|t|}$: $y(t)$ — $\frac{2\alpha}{\alpha^2 + (2\pi f)^2}$: $Y(f)$

$\frac{\alpha^2}{4\pi^2} = 1$

$\alpha^2 = 4\pi^2$

$\alpha = 2\pi$

$x(t) \longrightarrow X(f)$

$X(t) \longrightarrow x(-f)$

} Duality

$Y(t) = \frac{2\alpha}{4\pi^2} \cdot \frac{1}{\frac{\alpha^2}{4\pi^2} + t^2}$

$= \frac{1}{\pi} \left(\frac{1}{1+t^2} \right)$

$X(t) = \frac{1}{1+t^2} \xrightarrow{F} x(-f)$

$\frac{2\alpha}{4\pi^2} \quad \frac{1}{\frac{\alpha^2}{4\pi^2} + t^2} \xrightarrow{\mathcal{F}} e^{-\alpha|-f|} = e^{-\alpha|f|}$

$Y(t) = \frac{1}{\pi} \cdot x(t)$

$\frac{1}{\frac{\alpha^2}{4\pi^2} + t^2} \longrightarrow \frac{4\pi^2}{2\alpha} \cdot e^{-\alpha|f|}$

$\frac{\alpha^2}{4\pi^2} = 1 \qquad \alpha^2 = 4\pi^2 \rightarrow \alpha = 2\pi$

$\frac{1}{1+t^2} \xrightarrow{\mathcal{F}} \pi \cdot e^{-2\pi|f|}$

14) $x(t) = \frac{1}{1+t^2}$ Find the Fourier Transform. $X(f) = \int_{T_0} x(t) e^{-\cdots} dt$

$y(t) \quad Y(t)$

$e^{-\alpha|t|} \xrightarrow{F} \frac{2\alpha}{\alpha^2 + (2\pi f)^2} = \frac{2\alpha}{4\pi^2} \cdot \frac{1}{\frac{\alpha^2}{4\pi^2} + f^2}$ $\quad y(-f)$

$Y(t) = \frac{2\alpha}{\alpha^2 + (2\pi f)^2}$

$Y(f) = \frac{2\alpha}{\alpha^2 + (2\pi f)^2}$

$x(t) \longrightarrow X(f)$

$X(t) \longrightarrow x(-f)$ } Duality

$X(t) = \frac{1}{1+t^2} \xrightarrow{F} x(-f)$

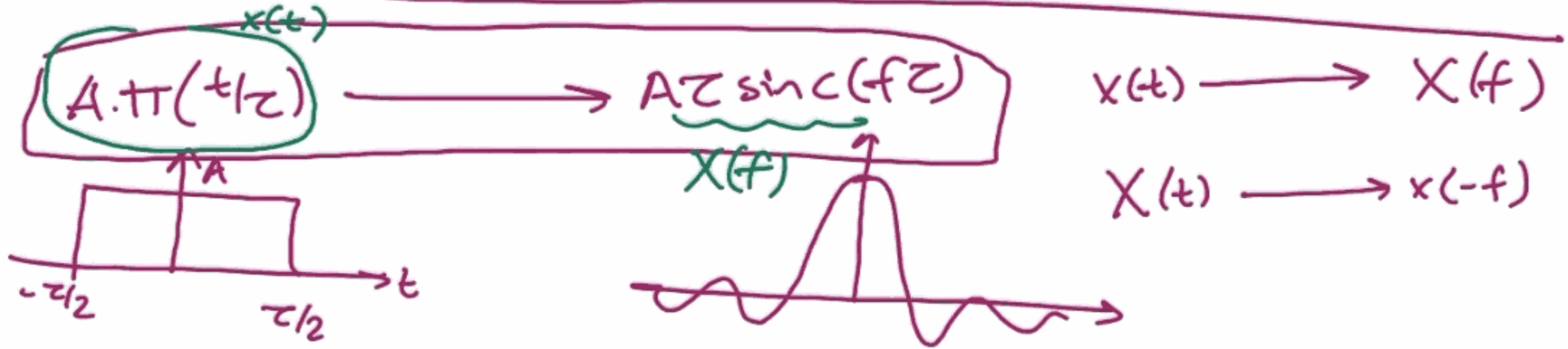
$\frac{2\alpha}{4\pi^2} \quad \frac{1}{\frac{\alpha^2}{4\pi^2} + t^2} \xrightarrow{F} e^{-\alpha|-f|} = e^{-\alpha|f|}$

$\frac{1}{\frac{\alpha^2}{4\pi^2} + t^2} \longrightarrow \frac{4\pi^2}{2\alpha} \cdot e^{-\alpha|f|}$

$\frac{2\alpha}{\alpha^2 + (2\pi t)^2} \xrightarrow{F} e^{-\alpha|-f|} = e^{-\alpha|f|}$

$\frac{\alpha^2}{4\pi^2} = 1 \qquad \alpha^2 = 4\pi^2 \rightarrow \underline{\alpha = 2\pi}$

$\frac{1}{1+t^2} \xrightarrow{F} \pi \cdot e^{-2\pi|f|}$



$x(t) \longrightarrow X(f)$

$X(t) \longrightarrow x(-f)$

Duality!

$A\tau \cdot \mathrm{sinc}(t\tau) \xrightarrow{\mathcal{F}} A \cdot \Pi(f/\tau)$

$$X(t) \longrightarrow x(-f) = A \cdot \Pi(-f/\tau) = A \cdot \Pi(f/\tau) \leftarrow$$

5. Aşağıdaki ifadenin eşitliği hangisidir? (5 Puan)

$\sqrt{2}e^{(j9\pi/4)}$ $= \sqrt{2}\left[\cos\left(\frac{9\pi}{4}\right) + j\sin\left(\frac{9\pi}{4}\right)\right]$

$e^{j\theta} = \cos\theta + j\sin\theta$

$= 1 + j$

- (●) $1 + j$
- ( ) $0.5j$
- ( ) $1 - j$
- ( ) $0.5 - 0.5j$

$a \cdot x(t) \xrightarrow{T} a \cdot y(t)$

2. y(t) ile verilen aşağıdaki sistem şıklarda verilen özeliklerden hangilerini sağlar? (8 Puan)

$y(t) = t^2 x(t-1)$

$x(t) \longrightarrow y(t) = t^2 \cdot x(t-1)$

$x_1(t) \longrightarrow y_1(t) = t^2 \cdot x_1(t-1)$

$x_2(t) = a \cdot x(t) \longrightarrow y_2(t) = t^2 \cdot a \cdot x(t-1)$

$T[ax(t)] = a \cdot T[x(t)]$

$= a \cdot t^2 \cdot x(t-1)$

- ○ *Lineer, zamanla* deg *ismez.*
- ○ *Lineer* deg *ildir, zamanla* deg *isir.*
- ○ *Lineer* deg *ildir, zamanla* deg *ismez.*
- ○ *Lineer, zamanla* deg *isir.*

$a \cdot x(t) \longrightarrow a \cdot y(t)$

2. y(t) ile verilen aşağıdaki sistem şıklarda verilen özeliklerden hangilerini sağlar? (8 Puan)

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$x_2(t) = a \cdot x(t) \longrightarrow y_2(t) = t^2 \cdot a \cdot x(t-1)$

$T[ax(t)] = a \cdot T[x(t)]$

$= a \cdot t^2 \cdot x(t-1)$

$x_2(t) \longrightarrow y_2(t) = t^2 \cdot x_2(t-1)$

$x_2(t) = x(t-t_0)$ $\quad y_2(t) = t^2 \cdot x(t-t_0-1)$ $\neq$

$y(t-t_0) = (t-t_0)^2 x(t-t_0-1)$

$x(t-t_0) \rightarrow y(t-t_0)$

- ○ *Lineer, zamanla* deg *ismez.*
- ○ ~~*Lineer deg ildir, zamanla* deg *isir.*~~
- ○ ~~*Lineer deg ildir, zamanla* deg *ismez.*~~
- ○ *Lineer, zamanla* deg *isir.*

$x[n] * \delta[n-n_0] = x[n-n_0]$, $x[n] \cdot \delta[n-n_0] = x[n_0]\,\delta[n-n_0]$

4. ~~Dürtü cevabı h[n]= -δ[n]+δ[n+2]~~ olan sistemin x[n]=u[n-3] giriş işareti için elde edilecek çıkış işaretini bulunuz. (10 Puan)

$x[n] \rightarrow [\,h[n]\,] \rightarrow y[n] = x[n] * h[n]$

- ◯ $-u[n-3]-u[n-2]$
- ◯ $-\delta[n]+\delta[n+2]$
- ◯ $u[n-3]$
- ◯ $\delta[n-1]+\delta[n-2]$

$y[n] = (u[n-3]) * (-\delta[n] + \delta[n+2])$

$= -\delta[n] * u[n-3] + \delta[n+2] * u[n-3]$

$= -u[n-3] + u[n+2-3]$

$= u[n-1] - u[n-3]$

